

//Count the len of the string or word.

Input : gelly

Output: 5

```
#include <stdio.h>
```

```
#define MAX_SIZE 100
```

```
#include <string.h>
```

```
int main() {  
    char str[MAX_SIZE];  
    scanf("%[^\\n]",str);  
    int len = strlen(str);  
    printf("Length of the string is %d",len);  
  
    return 0;  
}
```

//C program to copy one string to another string

Input : love

Output: love

```
#include <stdio.h>
```

```
#define MAX_SIZE 100
```

```
int main()
```

```
{
```

```
    char arr1[MAX_SIZE];
```

```
    char arr2[MAX_SIZE];
```

```
    int c=0,i;
```

```
    fgets(arr1,MAX_SIZE,stdin);
```

```
    for(i=0;arr1[i]!='\\0' ; i++){
```

```
        arr2[i] = arr1[i];
```

```
    }
```

```
    arr2[i] ='\\0';
```

```
    printf("%s", arr2);  
}
```

//concatenate two strings

Input : hi + hello

Output: hihello

```
#include <stdio.h>  
#include<string.h>  
#define MAX_SIZE 100  
int main()  
{  
    char arr1[MAX_SIZE];  
    char arr2[MAX_SIZE];  
    char arr3[MAX_SIZE];  
    int c=0,i,j,k;  
    fgets(arr1,MAX_SIZE,stdin);  
    fgets(arr2,MAX_SIZE,stdin);  
    for(i =0 ; arr1[i] != '\0' ; i++){  
        arr3[i] = arr1[i];  
  
    }  
    for(j= i-1,k=0 ; arr2[j] != '\0',arr2[k] != '\0' ;j++,k++){  
        arr3[j] = arr2[k];  
    }  
    printf("%s", arr3);  
    // printf("%s",strcat(arr1,arr2));  
  
}
```

/compare the string(correct way)

Input : gelly gelly

Output: 0

```
#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100
int main() {
    char arr1[MAX_SIZE];
    char arr2[MAX_SIZE];

    fgets(arr1, MAX_SIZE, stdin);
    fgets(arr2, MAX_SIZE, stdin);

    if (arr1[strlen(arr1) - 1] == '\n')
        arr1[strlen(arr1) - 1] = '\0';
    if (arr2[strlen(arr2) - 1] == '\n')
        arr2[strlen(arr2) - 1] = '\0';
    printf("%d", strcmp(arr1, arr2));
    return 0;
}
```

Another approach

```
#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100
int main() {
    char arr1[MAX_SIZE];
    char arr2[MAX_SIZE];
    scanf("%[^\\n]", arr1);
    scanf("%[^\\n]", arr2);
    int res = strcmp(arr1, arr2);
    printf("%d ", res);
}
```

```
    return 0;
}
```

Reversing a String...

Input : i am chellam

Output: chellam am i

```
#include <stdio.h>
#include <string.h>
int main()
{ char str[100];
  scanf("%[^\n]",str);
  int len = strlen(str);
  printf("%d\n",len);
  int word_index[50];
  int index=1;
  word_index[0]=0;
  char ch = ' ';
  for(int i=0; i<len; i++){
    if(str[i] == ch){
      word_index[index]=i+1;
      index++;
    }
  }
  for(int i=index-1; i>=0; i--){
    for(int j=word_index[i]; j<len; j++){
      if(str[j] != ch)
        printf("%c",str[j]);
      else
        break;
    }
    printf(" ");
  }
```

```
}  
}
```

//convert lower or upper

Input : gelly

Output: GELLY

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define MAX_SIZE 100
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    char arr1[MAX_SIZE];
```

```
    char arr2[MAX_SIZE];
```

```
    fgets(arr1, MAX_SIZE, stdin);
```

```
    fgets(arr2, MAX_SIZE, stdin);
```

```
    for(int i=0 ;arr1[i] != '\0'; i++){
```

```
        // printf("%c",tolower(arr1[i]));
```

```
        printf("%c",toupper(arr1[i]));
```

```
    }
```

```
}
```

//print string as togglecase

Input : GeLLy

Output: gELlY

```
#include <stdio.h>
```

```
#define MAX_SIZE 100
```

```
#include<stdlib.h>
```

```
#include<string.h>
```

```
int main()
```

```
{
```

```
    char str1[MAX_SIZE];
```

```

scanf("%[^\\n]", str1);
int len = strlen(str1);
int word_index[MAX_SIZE];
word_index[0]= 0;
int index= 1;
char ch = ' ';
for( int i=0 ; i< len ; i++){
    if(str1 [i] == ch){
        word_index[index] = i+1;
        index++;
    }
}

for(int i= index-1; i>=0 ;i--){
    for( int j=word_index[i] ; j<len ; j++){
        if(str1[j] != ch){
            printf("%c", str1[j]);
        }
        else
            break;
    }
    if(i!=0)
        printf(" ");
}

}

```

// Palindrome of a string

Input : MOM

Output: MOM

```
#include <stdio.h>
#include <string.h>
```

```
int main()
{
    char str[100];
    scanf("%[^\n]",str);
    int len = strlen(str);
    char rev[100];
    int index=0;
    for(int i=len-1;i>=0; i--){
        rev[index]=str[i];
        index++;
    }
    int res = strcmp(str,rev);
    if(res==0){
        printf("Palindrome");
    }
    else{
        printf("Not a Palindrome");
    }
}
```

// Find first occurrence index of the character

Input : gelly l

Output: index:2

```
#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100
```

```

int main()
{
    char str[MAX_SIZE];
    fgets(str,MAX_SIZE,stdin);
    char search = getchar();
    int len = strlen(str);
    int index=-1;
    for(int i=0; i< len ; i++){
        if(str [i] == search){
            index=i;
            break;
        }
    }
    if( index == -1){
        printf(" character not found");
    }
    else{
        printf("%c found at %d", search,index);
    }
}

```

// Find last occurrence index of the character

Input : gelly l

Output: index :3

```

#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100

```

```

int main()
{

```



```

char str[MAX_SIZE];
fgets(str,MAX_SIZE,stdin);
char search = getchar();
int len = strlen(str);
int index=-1;
for(int i=len-1; i>=0; i--){
    if(str [i] == search){
        index=i;
        break;
    }
}
if( index == -1){
    printf(" character not found");
}
else{
    printf("%c found at %d", search,index);
}
}

```

// Find all occurrences of a char

Input : gelly l

Output: index: 2 3

```

#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100

```

```

int main()
{
    char str[MAX_SIZE];
    fgets(str,MAX_SIZE,stdin);

```

```

char search = getchar();
int len = strlen(str);
int index=-1;
for(int i=0; i< len; i++){
    if(str [i] == search){
        index= i;
        printf("%c found at %d", search,index);
        printf("\n");

    }
}
if( index == -1){
    printf(" character not found");
}
}

```

//Count all occurrences

Input : gelly l

Output: 2

```

#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100

int main()
{
    char str[MAX_SIZE];
    fgets(str,MAX_SIZE,stdin);
    char search = getchar();
    int len = strlen(str);
    int c=0;
    for(int i=0; i< len-1; i++){
        if(str [i] == search){

```

```

        c++;
    }
}
printf("%d", c);
}

```

//find frequency

Input : gelly

Output: idhu ungaluke therui..ha ha haa

```

#include <stdio.h>
#include <stdlib.h>
int main() {
    char arr[100];
    int freq[100];
    scanf("%[^\n]", &arr);
    int len = strlen(arr);
    for(int i=0; i<len; i++){
        freq[i] = -1;
    }
    for(int i=0; i<len; i++){
        int count=1;
        for(int j=i+1; j<len; j++){
            if(arr[i]==arr[j]){
                count++;
                freq[j]=0;
            }
        }
        if(freq[i]!=0){
            freq[i]=count;
        }
    }
    for(int i=0 ; i<len ;i++){

```

```

        printf("%d ",freq[i]);
    }
    printf("\n");
    for(int i=0; i<len ;i++){
        if(freq[i]!=0 && arr[i] != ' ')
            printf("%c occurred %d times \n",arr[i],freq[i]);
    }
    return 0;
}

```

//find high frequency

```

#include <stdio.h>
#include <stdlib.h>
#include<string.h>
int main() {
    char arr[100];
    int freq[100];
    int i;
    int index=0;
    scanf(" %[^\n]",arr);
    int len = strlen(arr);
    for(int i=0; i<len; i++){
        freq[i]= -1;
    }
    for( i=0; i<len; i++){
        int count=1;
        for(int j=i+1; j<len; j++){
            if(arr[i]==arr[j]){
                count++;
                freq[j]=0;
            }
        }
    }
}

```

```

    }
    if(freq[i]!=0){
        freq[i]=count;
    }
}
for( i=0 ; i<len ;i++){
    printf("%d ",freq[i]);
}
printf("\n");
for(i=0; i<len ;i++){
    if(freq[i]!=0 && arr[i] != ' ')
        printf("%c occurred %d times \n",arr[i],freq[i]);
}
int max = freq[0];
for( i=0 ;i< len ; i++){
    if(freq[i] > max){
        max= freq[i];
        index=i;
    }
}
printf("%c occurs max times - %d", arr[index], max);
return 0;
}

```

//delete first occurrence

Input : kanithachellam

Output: kaithachellam

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```

char str[100];
scanf( " %[^\\n]",str);
int len= strlen( str);
char letter;
scanf("%c",&letter);
printf("%c\\n",letter);
int index=-1;
for( int i =0 ; i< len ; i++){
    if( str[i] == letter ){
        index =i;
        break;
    }
}
if (index != -1){
    for( int j =0 ; j< len ; j++){
        if (index == j){
            continue;
        }
        else
            printf("%c",str[j]);
    }

}
else
    printf("letter not found");
}

```

// delete last occurrence

Input : kanithachellam

Output: kaithachellm

```
#include <stdio.h>
```

```

#include <string.h>
int main()
{
    char str[100];
    scanf( " %[^\\n]",str);
    int len= strlen( str);
    char letter;
    scanf("%c",&letter);
    printf("%c\\n",letter);
    int index=-1;
    for( int i =len-1 ; i>=0 ; i--){
        if( str[i] == letter ){
            index =i;
            break;
        }
    }
    if (index != -1){
        for( int j =0 ; j <len ; j++){
            if (index == j){
                continue;
            }
            else
                printf("%c",str[j]);
        }
    }
    else
        printf("letter not found");
}

```

//delete all occurrences of a character

Input : kanithachellam

Output: kithchellm

```

#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#define MAX_SIZE 100 // Maximum string size
int main()
{
    char str[MAX_SIZE];
    char word;
    scanf("%[^\n]", str);
    scanf("%c", &word);
    int len = strlen(str);
    int l;
    for( int i=0 ;i<len ; i++){
        if(str[i] == word){
            int k = i;
            for( int j=k; j < len-1 ; j++){
                str[j] =str [j+1];

            }
            len--;
            i--;

        }
    }
    for( l=0 ; l< len ; l++){
        printf("%c",str[l]);

    }
}

```

//delete the duplicates

Input : gelly

Output: gely

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    char arr[100];
    int freq[100];
    scanf("%s",&arr);
    int len = strlen(arr);
    for(int i=0; i<len; i++){
        freq[i]= -1;
    }
    for(int i=0; i<len; i++){
        int count=1;
        for(int j=i+1; j<len; j++){
            if(arr[i]==arr[j]){
                count++;
                freq[j]=0;
            }
        }
        if(freq[i]!=0){
            freq[i]=count;
        }
    }
    for(int i=0 ; i<len ;i++){
        printf("%d ",freq[i]);
    }
    printf("\n");
    for(int i=0; i<len ;i++){
        if(freq[i] != 0 && arr[i] != ' ')
            printf("%c", arr[i]);
    }
}
```

```
    }  
    return 0;  
}
```

//replace the first occurrence

Input : g.ell . y (old - .) (new - !)

Output: g!ell.y

```
#include <stdio.h>  
#include <stdlib.h>  
#include <string.h>  
int main()  
{  
    char str[100];  
    char old;  
    char neww;  
    scanf("%[^\\n]",str);  
    int len = strlen(str);  
    scanf("%c", &old);  
    scanf("%c", &neww);  
    for( int i=0 ; i< len ; i++){  
        if( str [i] == old){  
            str[i] = neww;  
            break;  
        }  
    }  
    printf("%s",str);  
}
```

//replace the last occurrence

Input : g.ell . y (old - .) (new - !)

Output: g.ell!y

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main()
{
    char str[100];
    char old;
    char neww;
    scanf("%[^\\n]",str);
    int len = strlen(str);
    scanf(" %c", &old);
    scanf(" %c", &neww);
    for( int i=len-1 ; i>=0 ; i--){
        if( str [i] == old){
            str[i] = neww;
            break;
        }
    }
    printf("%s",str);
}
//replace the last occurrence
Input : g.ell . y   (old - . ) (new - ! )
Output: g!ell!y
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main()
{
    char str[100];
    char old;
    char neww;
```

```

scanf("%[^\\n]",str);
int len = strlen(str);
scanf(" %c", &old);
scanf(" %c", &neww);
for( int i=0; i< len ; i++){
    if( str [i] == old){
        str[i] = neww;

    }
}
printf("%s",str);
}

```

//Trim leading white space

Input : hiii

Output: hii

// Trim trailing white space

Input : hiii .

Output: hii

```

#include <stdio.h>
#define MAX_SIZE 100
#include <string.h>

```

```

int main()
{
    char str[MAX_SIZE];
    scanf("%[^\\n]",str);
    int len= strlen(str);
    int end = len-1;
    while(str[end]!=' '){
        end--;
    }
}

```

```

int start =0;
while(str[start]==' '){
    start++;
}
printf("\n");
for(int i=start; i<=end; i++){
    printf("%c",str[i]);
}
printf("\n");
}

```

// finding the all occurrence of a word in a string
 // find first and last found also

```

#include <stdio.h>
#define MAX_SIZE 100
#include <string.h>
int main()
{
    char str1[MAX_SIZE];
    scanf("%[^\n]",str1);
    char str2[MAX_SIZE];
    scanf("%[^\n]",str2);
    int len = strlen(str1);
    int len2 = strlen(str2);
    str2[len2]='\0';
    int word_index[MAX_SIZE];
    word_index[0]=0;
    int index=1;
    for(int i=0; i<len; i++){

```

```

    if(str1[i]==' '){
        word_index[index]=i+1;
        index++;
    }
}
for(int i=0; i<index; i++){
    printf("%d ",word_index[i]);
}
printf("\n");
int occurrence_arr[MAX_SIZE];
int occurrence_index=0;
for(int i=0; i<index; i++){
    int start=word_index[i];
    int flag=0;
    int k=0;
    int word_end=0;
    for(int j=word_index[i]; j<start+len2; j++){
        if(str1[j]!=str2[k]){
            flag=1;
            break;
        }
        if(j==(start+len2-1)){
            if((str1[j+1]==' ')||(str1[j+1]=='\0')){
                word_end=1;
            }
        }
        k++;
    }
    if((flag==0)&&(word_end==1)){
        occurrence_arr[occurrence_index]=word_index[i];
        occurrence_index++;
    }
}

```

```

}
if(occurance_index==0){
    printf("Word Not Found");
}
else{
    for( int i=0 ; i< occurance_index ; i++){
        printf("%d ", occurance_arr[i]);
    }
}
}

```

/// Removal of first and last and all occurrence of a word in a string

```

#include <stdio.h>
#define MAX_SIZE 100
#include <string.h>
int main()
{
    char str1[MAX_SIZE];
    scanf(" %[^\\n]",str1);
    char str2[MAX_SIZE];
    scanf(" %[^\\n]",str2);
    int len = strlen(str1);
    int len2 = strlen(str2);
    str1[len]=' ';
    str1[len+1]='\\0';
    str2[len2]='\\0';
    int word_index[MAX_SIZE];
    word_index[0]=0;
    int index=1;
    for(int i=0; i<len; i++){
        if(str1[i]==' '){

```

```

    word_index[index]=i+1;
    index++;
}
}
int occurrence_arr[MAX_SIZE];
int occurrence_index=0;
for(int i=0; i<index; i++){
    int start=word_index[i];
    int flag=0;
    int k=0;
    int word_end=0;
    for(int j=word_index[i]; j<start+len2; j++){
        if(str1[j]!=str2[k]){
            flag=1;
            break;
        }
        if(j==(start+len2-1)){
            if((str1[j+1]==' ')||(str1[j+1]=='\0')){
                word_end=1;
            }
        }
        k++;
    }
    if((flag==0)&&(word_end==1)){
        occurrence_arr[occurrence_index]=word_index[i];
        occurrence_index++;
    }
}
for(int i=0; i<occurrence_index; i++){
    printf("%d ",occurrence_arr[i]);
}
printf("\n");

```



```
if(occurance_index==0){
    printf("Word Not Found");
}
else{
printf("\nRemoval of First Occurance:\n");
for(int i=0; i<len; i++){
    if(i==occurance_arr[0]){
        i=i+len2;
    }
    else{
        printf("%c",str1[i]);
    }
}
printf("\nRemoval of last Occurance:\n");
for(int i=0; i<len; i++){
    if(i==occurance_arr[occurance_index-1]){
        i=i+len2;
    }
    else{
        printf("%c",str1[i]);
    }
}
printf("\nRemoval of all Occurance:\n");
for(int i=0; i<len; i++){
    int flag=0;
    for(int j=0; j<occurance_index; j++){
        if(i==occurance_arr[j]){
            i=i+len2;
            flag=1;
            break;
        }
    }
}
```

```
        if(flag==0)
            printf("%c",str1[i]);
    }
}
}
```

//To print lowercase in uppercase and viceversa

Input: HeLlo Iam

Output: hEILO iAM

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str[100];
    scanf(" %[^\\n]",str);
    int len=strlen(str);
    for(int i=0; i<len; i++){
        if(str[i] >= 'a' && str[i]<='z'){
            printf("%c",toupper(str[i]));
        }
        else if(str[i] >= 'A' && str[i]<='Z'){
            printf("%c",tolower(str[i]));
        }
        else{
            printf("%c",str[i]);
        }
    }
}
```

/// Anagram

```
#include <stdio.h>
```

```
#define MAX_SIZE 100
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char str1[MAX_SIZE];
```

```
    scanf("%[^\n]",str1);
```

```
    int len1 = strlen(str1);
```

```
    char str2[MAX_SIZE];
```

```
    scanf("%[^\n]",str2);
```

```
    int len2 = strlen(str2);
```

```
    if(len1!=len2){
```

```
        printf("This is Not an Anagram");
```

```
    }
```

```
    else{
```

```
        for(int i=0; i<len1; i++){
```

```
            str1[i] = tolower(str1[i]);
```

```
            str2[i] = tolower(str1[i]);
```

```
        }
```

```
        for(int i=0; i<len1; i++){
```

```
            for(int j=i+1; j<len1; j++){
```

```
                if(str1[i]>str1[j]){
```

```
                    char a = str1[i];
```

```
                    str1[i] = str1[j];
```

```
                    str1[j] = a;
```

```
                }
```

```
                if(str2[i]>str2[j]){
```

```
                    char a = str2[i];
```

```
                    str2[i] = str2[j];
```

```

        str2[j] = a;
    }
}
}
int res = strcmp(str1,str2);
if(res==0){
    printf("Its a Anagram");
}
else{
    printf("Not a Anagram");
}

}
}

```

// removal of irrelevant spaces in a String...

```

#include <stdio.h>
#include <string.h>
#define MAX_SIZE 100
int main()
{
    char str[MAX_SIZE];
    scanf("%[^\n]",str);
    int len = strlen(str);
    /// trimming and trailing the spaces
    int start =0;
    while(str[start]==' '){
        start++;
    }
    int end = len-1;
    while(str[end]==' '||str[end]=='\n'){

```

```
        end--;  
    }  
    ///print the string by removing extra spaces  
    printf("\n");  
    for(int i=start; i<=end; i++){  
        if(str[i]!=' '){  
            printf("%c",str[i]);  
        }  
        else{  
            printf(" ");  
            int temp = i;  
            while(str[temp]==' '){  
                temp++;  
            }  
            i=temp-1;  
        }  
    }  
    printf("\n");  
}
```


